# REVE EEG Experiment Report

Workspace summary generated from completed code and reports

April 28, 2026

**Abstract**

This document summarizes the completed REVE EEG experiments in the workspace. The experiments extract frozen REVE latents from the PhysioNet EEG Motor Movement/Imagery dataset, evaluate several linear-probe protocols, and analyze the low-rank geometry of the resulting 512-dimensional representations. The main empirical story is that simple pooled embeddings with PCA do not reproduce the paper-level result, while the paper-style flattened token head performs strongly. At the same time, spectrum and subspace experiments show that the representation is highly concentrated in a small number of directions, allowing a K=50 subspace probe to preserve much of the task signal.

## 1 Workspace And Data Context

The project is organized as a script-based research workspace. Experiment code lives in `code/`, text reports and logs live in `reports/`, and extraction logs live under `latents/`, `latents_pp/`, `latents_pp2/`, and `latents_paper/`. Several scripts still reference `/media/alex/DATA1/REVE/`; the current workspace is `/media/alex/DATA/REVE/`. The available completed summaries in this workspace are text reports rather than the original heavy HDF5/NPZ artifacts.

The dataset is PhysioNet EEGBCI. The task classes are left hand (L), right hand (R), both fists (B), and both feet (F). The paper-aligned extraction uses 108 subjects. The probe log reports 9,723 motor trials with nearly balanced class counts: 2,450 L, 2,410 R, 2,436 B, and 2,427 F.

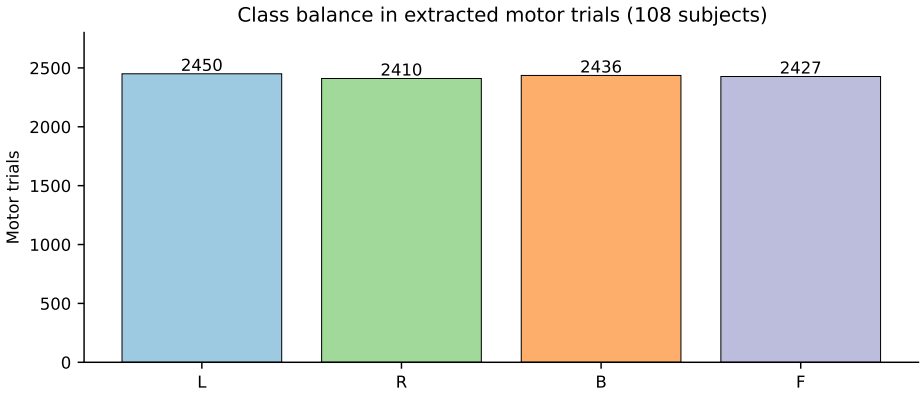


Figure 1: Class balance for the extracted motor trials used by the downstream probe experiments. The classes are close enough in size that raw accuracy and balanced accuracy are similar in most fixed-split experiments.

The canonical HDF5 layout from `code/extract_latents.py` stores continuous EEG, trial metadata, labels, task type, and REVE latents. For the paper-style experiments, each trial is represented by 64 channels and four patches per channel, producing 256 token embeddings of dimension 512.

## 2 Completed Experiment Pipeline

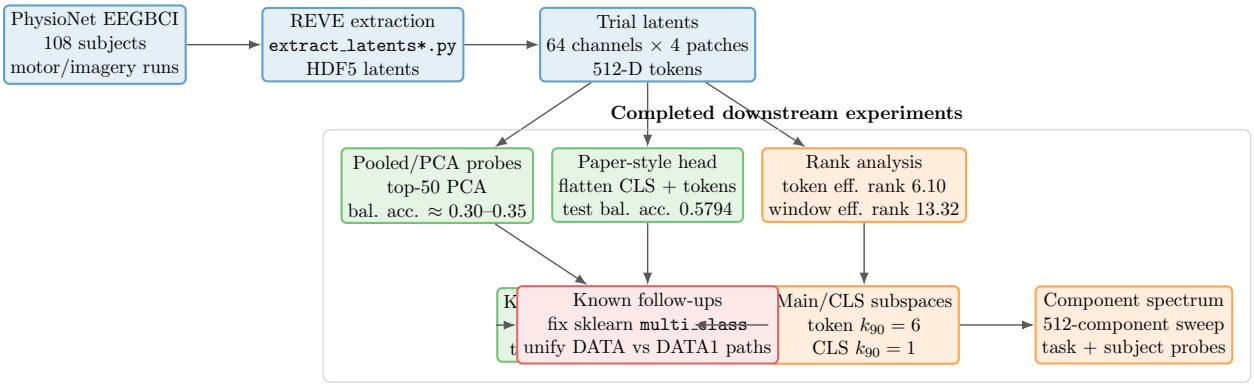


Figure 2: Completed experiment flow. The work branches into naive pooled probes, paper-protocol probing, and representation-geometry analyses that motivate subspace and component-level experiments.

## 3 Experiment 1: Latent Extraction

The extraction scripts load EEGBCI EDF runs, standardize MNE channel names, resample to 200 Hz, segment four-second windows, and pass each window through the frozen REVE model. The main extraction variants are:

- `extract_latents.py`: baseline latent extraction.
- `extract_latents_pp.py`: preprocessed extraction variant.
- `extract_latents_pp2.py`: global-zscore preprocessed variant.
- `extract_latents_paper.py`: paper-aligned extraction used by the strongest probe and geometry experiments.

The extraction logs indicate that all 108 subjects were processed. A typical subject contributes roughly 180 trials across the EEGBCI task runs. The latent tensor used by downstream scripts has shape $(N, 64, 4, 512)$, where the middle dimensions correspond to channels and patches.

## 4 Experiment 2: Pooled Linear Probes

The first probe family tests whether a simple pooled representation is already sufficient for four-way motor classification. These scripts attention-pool or otherwise reduce each trial to a 512-dimensional

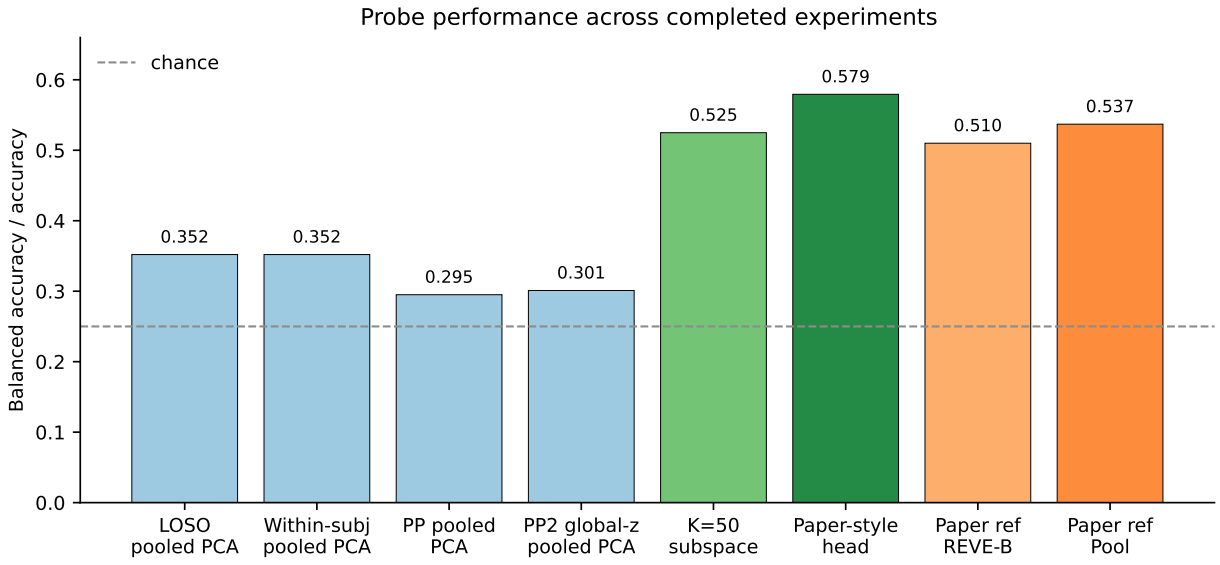


Figure 3: Classification results across completed probe experiments. The first four bars are pooled/PCA baselines; the later bars show the K=50 subspace probe, the paper-style flattened head, and the paper reference numbers.

vector, apply top-50 PCA components, and train a linear classifier. This family includes LOSO, within subject, preprocessed, and global-zscore variants.

The result is consistent across variants: pooled 512-dimensional features plus PCA do not explain the paper-level performance. The LOSO and within-subject reports both give mean accuracy 0.352. The preprocessed pooled variants are lower, with balanced accuracies 0.295 and 0.301. This suggests that the useful task information is not well preserved by a naive global pooling strategy, or that the paper protocol depends on retaining token-level structure.

## 5 Experiment 3: Paper-Protocol Head

The paper-style head in `code/train_paper_head.py` keeps the token structure instead of collapsing each trial to one vector. It learns a `cls_query_token`, computes an attention-pooled context vector, concatenates that context with the 256 token embeddings, flattens the resulting $(1 + 256) \times 512$ representation, and trains RMSNorm + dropout + a linear classifier.

The split is fixed by subject: subjects 1–70 for training, 71–89 for validation, and 90–109 for testing. The completed run reports 6,300 training trials, 1,620 validation trials, and 1,803 test trials. It reaches best validation balanced accuracy 0.5598 and test balanced accuracy 0.5794. This is above both cited paper references in the report: 0.510 for REVE-B and 0.537 for REVE-B with pooling.

The performance gap between pooled/PCA probes and the paper-style head is one of the main results of the workspace. It implies that token-level spatial and patch structure matters, even if the latent embedding dimension itself is strongly low-rank.

## 6 Experiment 4: Rank And Spectrum Analysis

The rank analyses ask how much of the 512-dimensional embedding space is actually used by the frozen REVE latents. The paper-preprocessed per-token analysis uses 2,489,088 token samples and finds effective rank 6.10. The per-window pooled analysis uses 9,723 windows and finds effective rank 13.32.

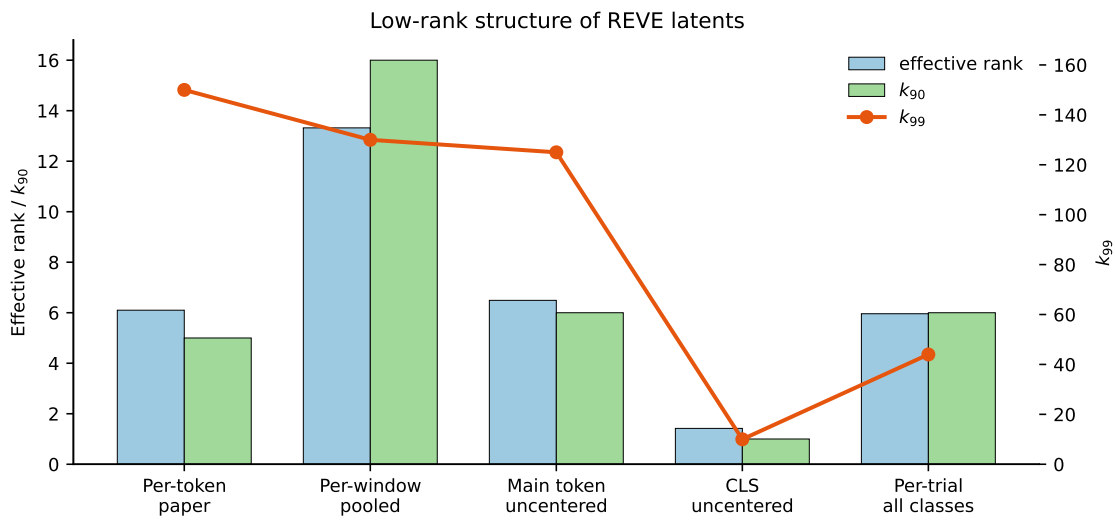


Figure 4: Effective rank, $k_{90}$, and $k_{99}$ across completed geometry analyses. The representation is highly concentrated: token-level energy reaches 90% in roughly five to six directions, and the CLS subspace reaches 90% in one direction.

| **Analysis** | **Eff. rank** | $k_{90}$ | $k_{99}$ |
|---|---|---|---|
| Paper latents, per-token samples | 6.10 | 5 | 150 |
| Paper latents, per-window pooled samples | 13.32 | 16 | 130 |
| Main token subspace, uncentered | 6.49 | 6 | 125 |
| CLS subspace, uncentered | 1.42 | 1 | 10 |
| Per-trial embeddings, all classes including rest | 5.96 | 6 | 44 |

Table 1: Summary of completed rank analyses. $k_{90}$ and $k_{99}$ denote the number of components needed to capture 90% and 99% of the reported spectrum.

The main token subspace report also shows that the global mean is not negligible: $\|\mu\|_2 = 51.6$, while $\sqrt{\lambda_1} = 72.3$. The CLS subspace is even more dominated by its first component, with $\|\mu_{CLS}\|_2 = 48.6$ and $\sqrt{\lambda_1} = 48.6$. This is important when interpreting uncentered eigendecompositions: the leading direction mixes representation variance with mean structure.

## 7 Experiment 5: Main And CLS Subspaces

The main subspace experiment computes an uncentered eigendecomposition of the per-token second moment. The CLS subspace experiment performs the analogous analysis on attention-pooled CLS vectors. The reports show sparse coordinate support among the top directions. For the top five token PCs, only 14 coordinates exceed absolute loading 0.1, and nine exceed 0.2. For the top five CLS PCs, 30 coordinates exceed 0.1, but only eight exceed 0.2.

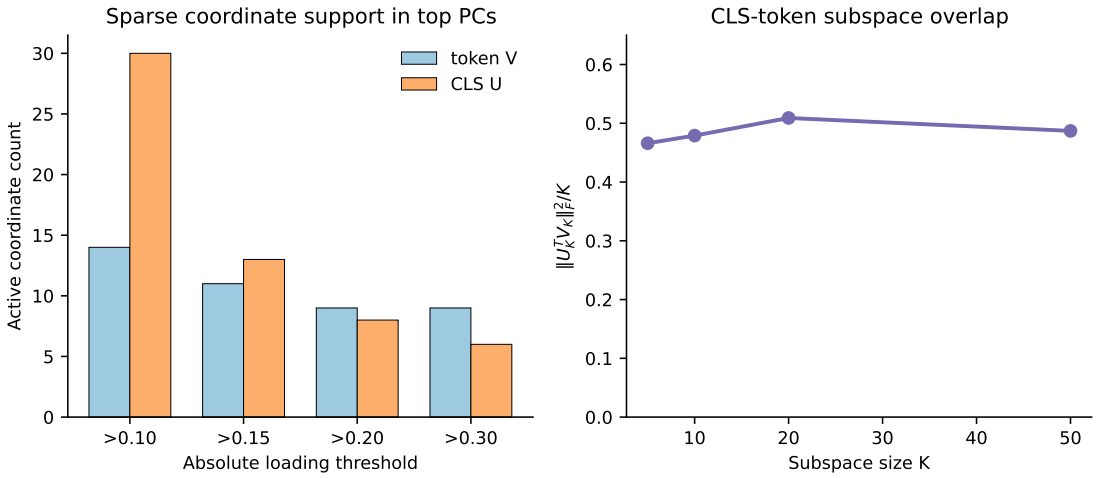


Figure 5: Left: active coordinate counts among the top token and CLS principal directions at different loading thresholds. Right: normalized subspace overlap between CLS subspaces $U_K$ and token subspaces $V_K$.

The CLS-token subspace overlap is moderate rather than complete: $\|U_5^T V_5\|_F^2/5 = 0.466$, rising to 0.509 at $K = 20$ and 0.487 at $K = 50$. Thus, the CLS and token summaries share substantial structure but are not interchangeable.

## 8 Experiment 6: K=50 Subspace Probe

The K=50 subspace probe projects the CLS representation into $U_{50}$ and all token embeddings into $V_{50}$, then concatenates the result. The feature dimension is $50 + 256 \times 50 = 12{,}850$, which is much smaller than the full flattened paper head with $(1 + 256) \times 512 = 131{,}584$ features.

The completed K=50 run reports:

- train/validation/test split: 6,300 / 1,620 / 1,803 trials;
- best validation balanced accuracy: 0.5161;
- test raw accuracy: 0.5247;
- test balanced accuracy: 0.5249.

This result is below the full paper-style head, but above the naive pooled/PCA probes and close to the cited paper REVE-B reference. It is the most useful intermediate representation in the completed experiments because it trades a large reduction in dimensionality for a modest performance loss and supports component-level interpretation.

## 9 Experiment 7: Component Spectrum

The full component-spectrum script computes all 512 token directions $V$ and all 512 CLS directions $U$, projects each trial into those bases, and sweeps single-component features. For each component index $k$, it records task balanced accuracy, subject balanced accuracy, and the corresponding token and CLS eigenvalues.

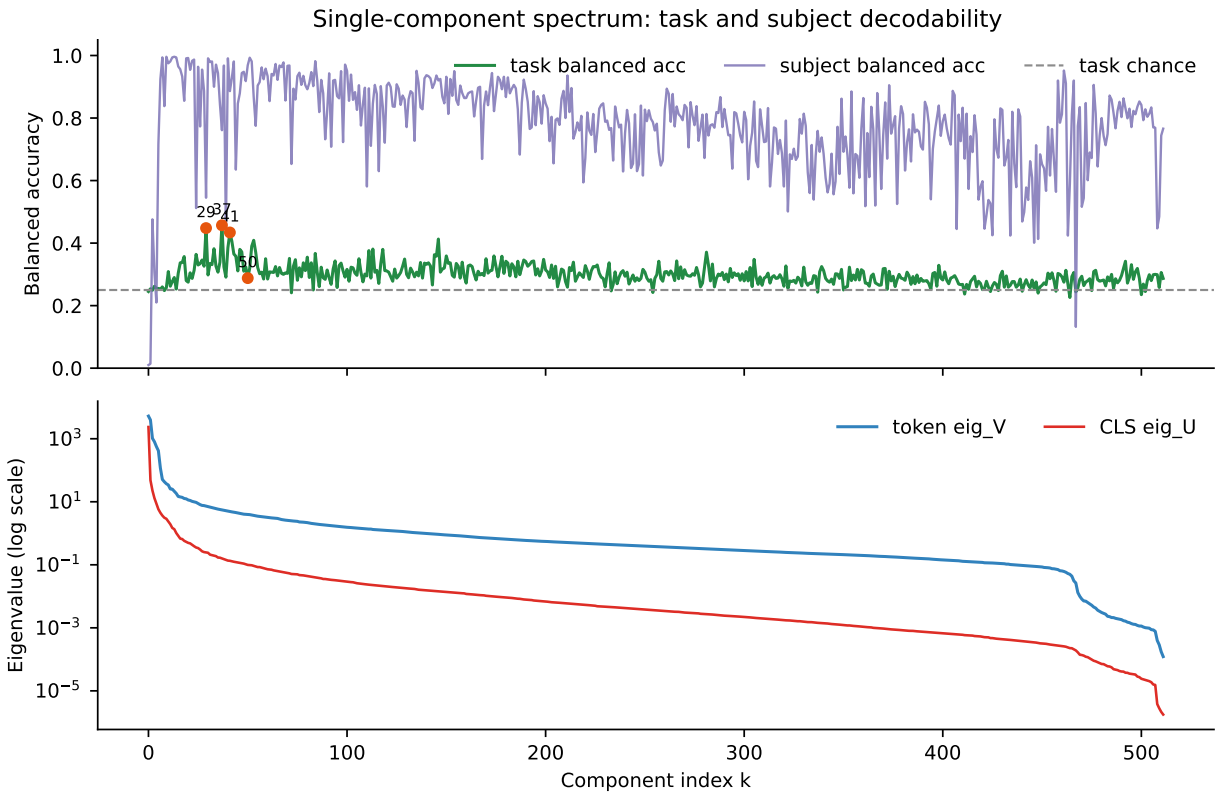


Figure 6: Full 512-component sweep. Top: task and subject balanced accuracy from single-component probes. Bottom: token and CLS eigenvalues on a log scale. The highlighted task-relevant components include $k = 29$, $k = 37$, and $k = 41$, which are also used by the Haufe-pattern script.

The spectrum experiment shows a sharp distinction between energy and task utility. The largest eigenvalue components are not the strongest task components: early components are dominated by global and subject-related structure, while task accuracy has local peaks deeper in the spectrum. For example, the text report shows task balanced accuracy 0.448 at $k = 29$, 0.457 at $k = 37$, and 0.434 at $k = 41$, even though the leading components carry much larger eigenvalues.

Subject identity is highly decodable from many individual components, often far above the task score. This supports the interpretation that subject-specific structure is a dominant axis of variation in the REVE latent space.

## 10 Experiment 8: Drop-K Study

The Drop-K experiment asks what happens when the largest directions are removed from the K=50 subspace features. If leading components encode global or subject-specific structure, removing them

may improve task discrimination.

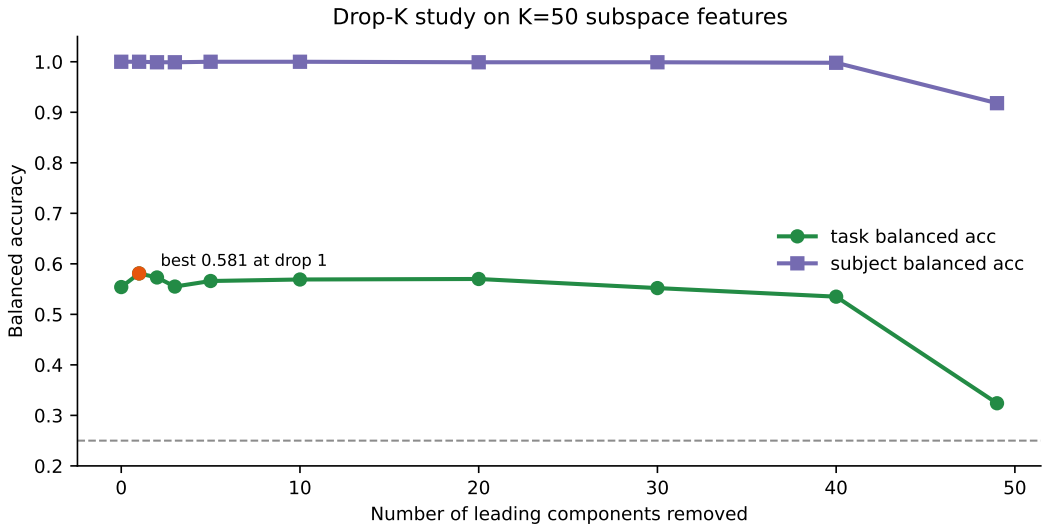


Figure 7: Drop-K results for K=50 subspace features. Removing the largest component improves task balanced accuracy from 0.554 to 0.581 in the completed report, while subject identity remains almost perfectly decodable for many settings.

The completed report confirms this intuition. With no dropped components, task balanced accuracy is 0.554 and subject balanced accuracy is 1.000. Dropping the top component improves task balanced accuracy to 0.581, while subject balanced accuracy remains 1.000. Even after dropping 20 leading components, task balanced accuracy remains 0.570 and subject balanced accuracy remains 0.999. Only near the extreme setting of dropping 49 components does task accuracy fall to 0.324.

## 11 Experiment 9: Haufe-Style Pattern Analysis

The Haufe-pattern scripts are designed to interpret the K=50 subspace probe. Instead of materializing the full 12,850 $\times$ 12,850 covariance matrix, the efficient version uses the fact that features in the $U_K$ and $V_K$ bases have diagonal covariance with entries equal to the corresponding eigenvalues. The pattern computation can therefore be written elementwise as

$$P[i, c] = \sigma_i W[c, i],$$

where $W$ is the classifier weight matrix and $\sigma_i$ is the eigenvalue associated with feature $i$.

The script then lifts token patterns back from the K-dimensional subspace into the original 512-dimensional REVE embedding space. Planned visualizations include per-class topomaps, top-task component topomaps for $k = 29$, $k = 37$, and $k = 41$, CLS pattern bars, and per-component pattern-energy summaries. No generated Haufe PDF is present in the current workspace, but the code path is set up to write `reve_haufe.pdf` when the required NPZ/HDF5 artifacts are available.

| Experiment | Feature/head | Reported result |
|---|---|---|
| LOSO linear probe | Top-50 PCA from 512-D pooled window embeddings | Mean accuracy 0.352 |
| Within-subject probe | 5-fold CV, top-50 PCA from pooled embeddings | Mean accuracy 0.352 |
| Preprocessed pooled probe | Attention-pooled 512-D features, top-50 PCA | Balanced accuracy 0.295 |
| Global-zscore pooled probe | Attention-pooled 512-D features, top-50 PCA | Balanced accuracy 0.301 |
| Paper-protocol head | Flattened CLS plus 256 token embeddings, RMSNorm, dropout, linear classifier | Test balanced accuracy 0.5794 |
| K=50 subspace probe | $U_K$(CLS) concatenated with $V_K$(tokens), 12,850-D features | Test balanced accuracy 0.5249 |
| Drop top component | K=50 subspace features after removing the largest direction | Task balanced accuracy 0.581 |

Table 2: Completed classification and probe experiments. The strongest completed result is the paper-style head, while the strongest compressed result is the Drop-K subspace variant.

## 12 Consolidated Results

## 13 Interpretation

The completed experiments support four main conclusions.

First, the naive pooled/PCA probe is not a faithful reproduction of the paper-style REVE evaluation. Across LOSO, within-subject, and preprocessing variants, pooled features land around 0.30–0.35 accuracy. This is only modestly above four-way chance and well below the paper references.

Second, preserving token structure is crucial. The paper-style head reaches 0.5794 balanced accuracy, showing that the extracted latents can support strong classification when the downstream head sees the full spatial-patch token layout.

Third, the latent space is highly anisotropic and low-rank. Token-level energy is concentrated in roughly five to six directions for 90% energy, and the CLS space is dominated by a single leading direction. This explains why subspace methods work, but it also warns that high-energy directions are not necessarily the most task-specific directions.

Fourth, task and subject information are entangled but separable enough to be studied component-wise. The component spectrum and Drop-K experiments show that subject identity is extremely easy to decode from many components, while useful task components can appear away from the very top of the eigenvalue spectrum.

## 14 Known Issues And Next Steps

- `reports/probe_run.log` shows a failed scikit-learn run: `LogisticRegression` rejected the `multi_class` keyword. The script should be updated for the installed scikit-learn version or the dependency should be pinned.
- Many scripts hard-code `/media/alex/DATA1/REVE/`. A shared configuration file or command-

line arguments would make reruns reproducible from the current `/media/alex/DATA/REVE/` workspace.

- The text reports reference heavy artifacts such as `reve_VU_full.npz`, `reve_proj_full.npz`, and HDF5 latent files, but these are not present under the current workspace path. Restoring or regenerating them is necessary for new Haufe/topomap figures.
- The next scientific step is to turn the component-spectrum and Drop-K observations into a controlled subject-invariance experiment: compare raw paper-head features, K=50 features, Drop-K features, and subject-residualized features under the same split.